

torch.geqrf#

<https://pytorch.org/docs/stable/generated/torch.geqrf.html#torch.geqrf>

Description:

This is a low-level function for calling LAPACK's `geqrf` directly. This function returns a namedtuple `(a, tau)` as defined in LAPACK documentation for `geqrf`. Computes a QR decomposition of input. Both Q and R matrices are stored in the same output tensor `a`. The elements of R are stored on and above the diagonal. Elementary reflectors (or Householder vectors) implicitly defining matrix Q are stored below the diagonal. The results of this function can be used together with `torch.linalg.householder_product()` to obtain the Q matrix or with `torch.ormqr()`, which uses an implicit representation of the Q matrix, for an efficient matrix-matrix multiplication. See LAPACK documentation for `geqrf` for further details. See also `torch.linalg.qr()`, which computes Q and R matrices, and `torch.linalg.lstsq()` with the `driver="gels"` option for a function that can solve matrix equations using a QR decomposition.

Function Signature

```
torch.geqrf(input, *, out=None)#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (tuple, optional) – the output tuple of (Tensor, Tensor). Ignored if None.
Default: None.

Notes & Warnings

NOTE

See also `torch.linalg.qr()`, which computes Q and R matrices, and `torch.linalg.lstsq()` with the `driver="gels"` option for a function that can solve matrix equations using a QR decomposition.

Detailed Documentation

Documentation

This is a low-level function for calling LAPACK's `geqrf` directly. This function returns a `namedtuple` (`a`, `tau`) as defined in LAPACK documentation for `geqrf`.

Computes a QR decomposition of input. Both `Q` and `R` matrices are stored in the same output tensor `a`. The elements of `R` are stored on and above the diagonal. Elementary reflectors (or Householder vectors) implicitly defining matrix `Q` are stored below the diagonal. The results of this function can be used together with `torch.linalg.householder_product()` to obtain the `Q` matrix or with `torch.ormqr()`, which uses an implicit representation of the `Q` matrix, for an efficient matrix-matrix multiplication.

See LAPACK documentation for `geqrf` for further details.

See also `torch.linalg.qr()`, which computes `Q` and `R` matrices, and `torch.linalg.lstsq()` with the `driver="gels"` option for a function that can solve matrix equations using a QR decomposition.

`input` (Tensor) – the input matrix

`out` (tuple, optional) – the output tuple of (Tensor, Tensor). Ignored if `None`. Default: `None`.